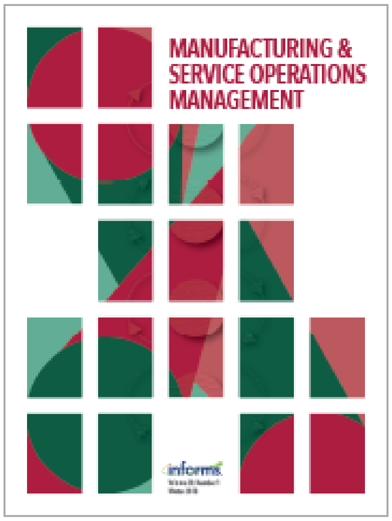

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Omnichannel Distribution to Fulfill Retail Orders

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Abstract. ***Problem definition*:** Although the analytical literature extensively studies distribution channels, empirical evidence on the value of omnichannel distribution is limited, especially for the omnichannel used by manufacturing companies to fulfill retail orders. I empirically evaluate the extent to which the dual–distribution center (dual-DC) distribution channel and the factory-direct distribution channel contribute to fulfilling orders from retail stores compared with the traditional single–distribution center (single-DC) channel. ***Academic/practical relevance*:** Many manufacturing companies develop their distribution omnichannel to fulfill retail orders. To make proper decisions on various channels, they need to understand the trade-offs between different order-fulfillment measures and costs in different distribution channels. ***Methodology*:** I exploit two switches in distribution channels of a manufacturing company to its retail customers: one from single-DC to dual-DC distribution and the other from single-DC to factory-direct distribution. To account for the trade-offs between order fulfillment and the costs associated with each distribution channel, I develop three equations for fill rate, lead time, and production and distribution costs in a difference-in-difference framework and then estimate the equations using proprietary data of retail orders and delivery records. ***Results*:** The results quantify the contributions of distribution channels to order fulfillment. Compared with the single-DC distribution channel, the dual-DC distribution channel raises the fill rate by 0.4% and reduces the lead time by 9.7% without incurring additional costs, whereas the factory-direct distribution channel increases the fill rate by 0.5% and provides a 5.2% cost savings but extends the lead time by 12.5%. I further analyze these contributions to order fulfillment across demand variability and order quantity. ***Managerial implications*:** The findings provide manufacturing companies with valuable knowledge of their distribution channel choices and means to find a cost-effective distribution channel to improve order fulfillment for various customers and products.

Supplemental Material: The online appendices are available at https://doi.org/10.1287/msom.2022.1104.

Keywords: distribution • omnichannel • order fulfillment • fill rate • lead time • retail orders

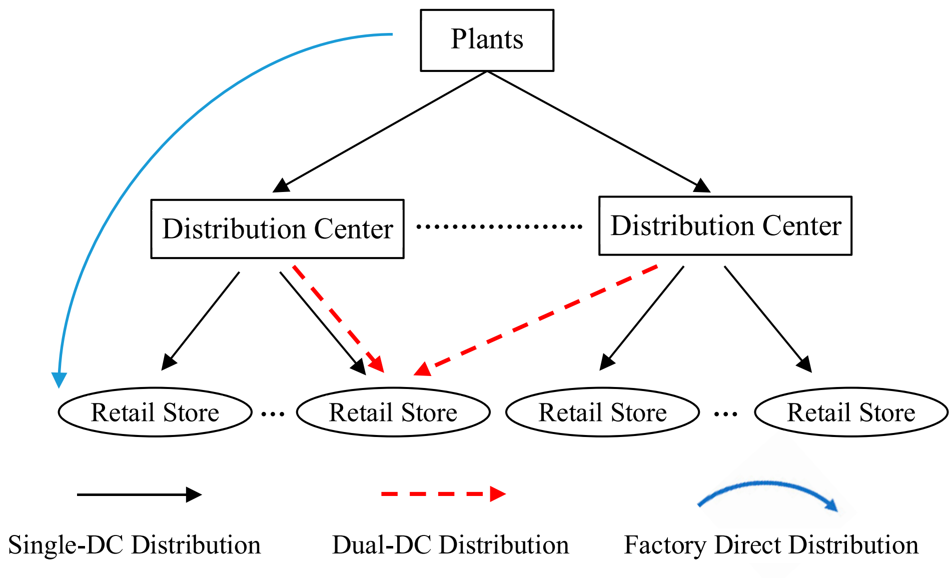
1. Introduction

Distribution channels are used in supply chains to fulfill customer orders. Whereas the literature mainly studies the delivery service provided by retailers to fulfill consumer orders (Cui et al. 2019, Fisher et al. 2019), there is limited understanding of the value of the distribution channels by which manufacturing companies fulfill retail orders. The existing knowledge of retailers' distribution channels to consumers may not be applied directly to manufacturing companies' distribution channels because retail orders aggregate consumer demand and manufacturers have larger supply capacities than retailers. I conduct this study to empirically quantify the contributions of distribution channels provided by manufacturing companies to retail order fulfillment as well as the trade-offs between the improved order fulfillment performance and costs.

Traditionally, in a consumer-packaged goods supply chain, manufacturing companies use their distribution centers to distribute products to retail stores. A retail store is commonly assigned to a distribution center, and orders from this retailer are fulfilled only by this distribution center (single-DC distribution). In addition to traditional single-DC distribution, I study two other distribution channels: dual-DC distribution, in which retail orders are fulfilled by two distribution centers, and factory-direct distribution, in which a production plant bypasses distribution centers and directly supplies retail stores. These three distribution channels are plotted in Figure 1.

Dual-DC distribution is a type of multiple-source distribution that provides more than one supply source and backup inventories to customers. Multiple sourcing is an efficient approach for pooling risk and providing a reliable supply (Agrawal et al. 2002, Allon and Van Mieghem 2010, Chen and Guo 2014). The theoretical value of factory-direct distribution is similar to direct store delivery in that it bypasses the

Figure 1. (Color online) The Omnichannel Used by the Manufacturer to Fulfill Retail Orders



distribution centers, thereby saving distribution costs. In 2008, the Grocery Manufacturer's Association reported that 24% of products were delivered directly to retail stores (bypassing the distribution centers) and these products accounted for 52% of profits (Nagle 2018). Whereas some companies (such as Honeywell and Nestle) bypass distribution centers, other companies, such as Kellogg, recently began moving away from the bypassing distribution center model toward the warehouse distribution model (Schroeder 2017). In order to find a cost-effective distribution channel and improve customer order fulfillment, it is important for manufacturing companies to understand the trade-offs between order fulfillment and costs associated with the different distribution channel options.

To explore the value of distribution channels, I worked with a U.S.-based production and distribution company of consumer packaged goods. This company uses its manufacturing factories to produce and distribution centers to deliver branded products to retail stores. The identification strategy is to exploit two changes in the distribution channels used by this company to fulfill orders from its retail customers. To explore the benefits of different distribution channels, the company switched distribution channels to some of its retail customers in 2012. First was a switch from single–distribution center (single-DC) distribution to dual–distribution center (dual-DC) distribution: in the 36th week of 2012, the company started using a second distribution center to provide additional supply to some retail stores that were supplied by single-DC distribution. The second distribution center serves as a backup source that delivers products to these stores when out of stock or low inventory occurs at the primary distribution center. After the switch, orders from these retail stores were fulfilled by the dual-DC distribution channel. Second, a switch from single-DC to factory-direct distribution: in the 21st week of 2012, the company reassigned some retail stores that were replenished by single-DC distribution to a manufacturing plant. After the switch, orders from these stores were fulfilled in a factory-direct distribution channel. The switches to dual-DC and factory-direct distribution provide opportunities to identify the value of these two channels.

A difference-in-difference (DiD) research design is developed to study these two switches in the distribution channels. Orders from retail stores and their fulfillment records were collected at the store–order–SKU level for 52 weeks in 2012. To provide "apple-to-apple" comparisons, I applied propensity score matching (PSM) to match stores in the dual-DC and factory-direct distribution channels with stores in the single-DC distribution channel (as the control group). I further apply the synthetic control method (SCM) (Abadie et al. 2010, 2015) as a robustness check for PSM.

I measure order fulfillment by the unit fill rate (in terms of availability) and lead time (in terms of speed). Because offering distribution channels generates costs, I also consider the production and distribution cost in distribution channels associated with manufacturing, inventory management, transportation, etc. The estimated results suggest that, as compared with the single-DC distribution channel, dual-DC distribution raises the fill rate by 0.4% and reduces lead time by 9.7% without increased costs, whereas factory-direct distribution increases the fill rate by 0.5%, extends the lead time by 12.5%, and reduces costs by 5.2%.

Although the literature suggests that delivery services provided by retailers may increase consumer demand (Cui et al. 2019, Fisher et al. 2019), the results in this paper indicate that dual-DC and factory-direct distribution do not result in increased order quantity or frequency from retail stores. Instead, this study suggests that these channels improve retail order fulfillment by altering the delivery quantity and frequency. Furthermore, I analyze these contributions to order fulfillment across products with various demand variability and order quantity. The findings reveal that high demand variability and order quantity amplifies these contributions by dual-DC and factory-direct distribution.

This paper contributes to the understanding of distribution channels provided by manufacturing companies. I investigate the trade-offs between order fulfillment and cost associated with distribution channels as well as how these trade-offs vary across products with various demand variability and order quantity. The results provide empirical evidence for the prescriptions from studies using analytical models and quantify theoretical effects associated with distribution channels. The findings provide manufacturing companies and retailers valuable knowledge of their distribution channel choices and means to find a cost-effective distribution channel to improve order fulfillment for various customers and products.

2. Literature Review

This study is related to the literature on distribution services, flexible supply, and order fulfillment. The existing studies suggest that service quality generates value for companies and their customers (Buell et al. 2016, Guajardo et al. 2016). In particular, the value of distribution and logistics services is explored by recent studies (Shi et al. 2013, Pendem and Deshpande 2020). Scholars find that distribution and logistics contribute to buyers' future behavior (Calvo et al. 2018), costs in supply chains (Shang et al. 2010), sales (Wan et al. 2018, Cui et al. 2019), and delivery performance (Peng and Lu 2017). This paper contributes to the literature on delivery services by investigating the outcomes of three distribution channels: single-DC, dual-DC, and factory-direct distribution.

The literature of sourcing suggests alternative supplies to companies for uncertainty mitigation in supply chains. In general, there are three supply structures: single, symmetric dual, and asymmetric dual supply (Sun and Li 2011, Li 2013). Dual/multiple supply sources are an efficient approach for pooling risk and providing reliable customer order fulfillment (Agrawal et al. 2002, Li and Debo 2009, Allon and Van Mieghem 2010). Chen and Guo (2014) find that, compared with a single supplier, multiple diversified suppliers help to reduce supply uncertainty. After analyzing a dual-supplier system of one responsible supplier with a high cost and a risky supplier with a low cost, Guo et al. (2016) suggest four strategies to use these two supplying sources. Kouvelis and Li (2008) find a backup supply in addition to the regular supply improves order fulfillment lead time. Tomlin (2006) compares the multiple-supplier approach with the holding-inventory approach in terms of managing supply risk and suggests a mixed mitigation strategy. To extend the literature that investigates the benefits of alternative supply (Tomlin and Wang 2005, Wang et al. 2010, Tang and Kouvelis 2011), I study two alternative distribution channels that provide flexible sources for manufacturing companies to fulfill customer orders. In particular, the dual-DC distribution channel provides products from a regular distribution center and a backup one, and the factory-direct distribution channel enables manufacturing plants to serve as an alternative source for customer orders.

Order fulfillment is generally evaluated by the fill rates (availability) and lead time (speed). For example, Özer and Xiong (2008) provide a study of how logistics postponement and fill rate affect a distribution channel's performance. Unit fill rate is a common measure of order fulfillment that indicates the inventory availability to retail orders (Wan et al. 2014, Craig et al. 2016, Zhou and Wan 2017). Additionally, it is generally accepted that reduction in lead time provides value to firms (Randall et al. 2006, Yao et al. 2009, Blackburn 2012, de Treville et al. 2014). Lead time reductions provide a shorter order fulfillment cycle and generate efficiency for buying firms. Shorter lead time helps reduce inventory costs and improve demand forecasting accuracy (Suri 1998, Zipkin 2000). This study empirically adds the consideration of cost to the order fulfillment literature that mainly assesses lead time and fill rate. I further study how the changes in fill rate, lead time, and costs resulting from the distribution channels vary across demand variability and order quantity.

In summary, the empirical analyses contribute to the literature by supporting the findings of previous modeling studies with empirical evidence and quantifying the theoretical effects associated with distribution channels. Whereas the literature provides theoretical prediction of the benefits of different distribution channels, the lack of empirical evidence and quantification of these effects limits our understanding. The empirical studies confirm the prescriptions from normative models and quantify the theoretical effects of distribution channels.

3. Empirical Context

3.1. Traditional Distribution Channel: Single-DC Distribution

I worked with a U.S. manufacturing and distribution company of consumer packaged goods that owns manufacturing plants and distribution centers. Within the company's distribution network, each retail store is assigned to a distribution center. The company makes the production plans and replenishes inventories in the distribution centers based on its forecasted retail orders. The company provided me with order records from its retail customers and deliveries.

In the distribution network of the company, plants produce and ship products to distribution centers that fulfill retail orders. Retail stores do not pay for the cost of distribution. Instead, the distribution company offers complimentary delivery to retail stores. Complimentary delivery to retail stores is common among manufacturing companies (such as Kraft Foods Group and Frito-Lay) in the consumer packaged goods industry. There are some advantages to maintaining full control of the distribution channel by providing complimentary deliveries. For example, they help companies minimize the distribution time of their products, especially for fresh products (such as milk and bread). Complimentary delivery also helps the manufacturing companies provide their products to small retailers, which do not have their own distribution channels.

The company employed a traditional distribution approach: the single-DC distribution channel, in which each distribution center supplies a group of retail stores based on their geographic locations. In the single-DC distribution channel, no store is supplied by more than one distribution center. Although order fulfillment priority may be applied in some industries (Alptekinoğlu et al. 2013), this company uses the first come, first served rule to fulfill retail orders. Order fulfillment decisions are made by the order dispatchers in the distribution centers. Based on the inventory in the distribution centers and the retail orders received, the order dispatchers allocate inventory and retail orders to truck drivers. On a daily basis, the dispatchers provide the truck drivers with a printed delivery store list, including order information and store addresses. The truck drivers load the ordered products into the trucks for deliveries the next morning. Throughout this study, the order fulfillment process remained unchanged in the plant and distribution centers included in the distribution channel switches.

If the inventory is insufficient to fulfill a retail order, the distribution center may either partially fulfill an order or postpone delivery to the store. In the case of partial fulfillment, the unit fill rate of the retail order becomes less than 100%. In the case of delayed delivery, the fulfillment lead time increases. There is no alternative supply source for stores in the single-DC distribution channel. In case of stockouts at its supplying distribution center, a retail store has to wait until the distribution center receives its incoming supply shipments. Some unsatisfied orders are aggregated to future orders from this retail order. These order aggregations result in the long lead time of retail orders.

3.2. Switches to Dual-DC and Factory-Direct Distribution

Since 2010, there has been increasing interest in a distribution omnichannel across many industries (Solomon 2015). Following this trend, the company endeavored to explore the potential benefits of alternative distribution methods. In 2012, the company made two changes to distribution channels for some retail stores. One is that, in week 36, the company allowed a distribution center as a backup supply source to some retail stores that were previously supplied by a single distribution center. After that, these retail stores were supplied by the dual-DC distribution channel. Retail orders were not split between two distribution centers, but were fulfilled by either the primary or the backup distribution center. The backup distribution center provides an alternative supply source to retail stores by shipping products to them when the primary distribution center is out of stock or does not have enough inventory. The other is that, in week 21 of 2012, the company reassigned some retail stores directly to a production plant.

These two distribution channel switches provide opportunities to identify the contribution of dual-DC and factory-direct distribution to retail order fulfillment. First, the switches to dual-DC and factory-direct distribution did not affect the company's product portfolio. Interviews with the company's management team confirmed that these three distribution channels were not used to distribute different product categories to stores. Second, the plant and backup distribution center did not make any changes to their transportation fleets before or after the two switches. No trucks, truck drivers, or additional inventories were added to the plant or distribution centers for these two switches. The changes in distribution channels provide direct influences on supply sources to retail orders. Third, these changes were exogenous to retail stores. The company did not communicate with retailers before the channel switches. When placing orders, retail managers did not necessarily know from where the incoming shipments came and had no information about the inventory level at their supply locations. Changes in distribution channels did not alter retailers' ordering decisions. In Section 8, I analyze the retail order quantity and frequency after the switches in the distribution channels and do not find significant changes in retail orders. Adding a backup distribution center and switching to factory-direct distribution allows me to compare these stores switching distribution channels with those stores that were consistently in the single-DC distribution channel and to compare the order fulfillment in the weeks before and after the channel switches happened.

4. Hypothesis Development

Following the literature, I measure order fulfillment by fill rate and lead time (Randall et al. 2006, Yao et al. 2009, Blackburn 2012, de Treville et al. 2014). Fill rate is the ratio of delivered quantity to order quantity, whereas lead time is the length of time between order placement and delivery.

4.1. Average Value of Dual-DC and Factory-Direct Distribution

In the make-to-stock industry, distribution centers manage the inventory based on the forecasting of retail demand. Because demand forecasting is not always accurate, a distribution center may not have enough inventory to satisfy retail orders all the time. As a result of the pooling effect, alternative sourcing helps to mitigate the uncertainties in supply chains (Yang et al. 2012, Chen and Guo 2014). The dual-DC distribution channel adds an alternative distribution center to supply retail stores. This additional distribution center fulfills retail orders when the primary distribution center cannot. This dual-DC distribution generates the risk-pooling effect (Kim and Benjaafar 2002, Benjaafar et al. 2005, Song and Zipkin 2009), which mitigates the uncertainties and better fulfills retail orders.

It is possible that the transportation time from the alternative distribution center in the dual-DC distribution channel to retail stores is longer than the transportation time from the primary distribution center to retail stores because the primary distribution center is normally located closer to these stores than the alternative distribution center. However, in the dual-DC distribution channel, the alternative distribution center is used to fulfill orders only when the primary distribution center is out of stock. In the case of out of stock, the lead time of waiting for supplies from the plant to the stockout primary distribution center and from this distribution center to retail stores is compared with the lead time of supplies from an alternative distribution center to retail stores. The former lead time in the single-DC distribution for the stockout distribution center includes the potential production time, transportation time from the plant to the stockout primary distribution center, and transportation time from the primary distribution center to retail stores. Meanwhile, the latter lead time for an alternative distribution center includes only the transportation time from the alternative distribution center to stores. The potential production time and the transportation time from the plant to the primary distribution center make the former lead time longer than the latter lead time. Thus, I propose that the dual-DC distribution channel reduces lead time.

Hypothesis 1. *Compared with the single-DC distribution channel, the dual-DC distribution channel provides a higher fill rate and a shorter lead time for retail orders.*

The factory-direct distribution channel allows production plants to bypass distribution centers and directly fulfill retail orders. Normally, a plant produces goods in large batches and supplies just a few distribution centers. The supply capability of a plant is much larger than a distribution center. Hence, given the larger supply capability associated with plants, the fill rate for retail orders in the factory-direct distribution channel is expected to be higher than that in the single-DC distribution channel.

Meanwhile, the retail order processing time at a plant is expected to be longer than the order processing time at a distribution center. The dispatching manager and inventory workers at a plant normally handle large delivery quantities and small delivery frequencies to distribution centers. Nevertheless, the deliveries to retail stores have fewer quantities and are more frequent than those to distribution centers. With no additional personnel added to the plant, it may take longer for a plant to process retail orders than a distribution center does.

Furthermore, the delivery services offered by a plant are not as flexible as those offered by a distribution center. Compared with a plant, a distribution center has a wider variety of delivery trucks and a larger group of drivers. The better equipped distribution fleet allows a distribution center to fulfill retail orders more flexibly and in a shorter time than a plant. Moreover, within the distribution companies, distribution fleets (trucks and drivers) are exclusively assigned to specific facilities (distribution centers and plants) and normally are not shared with or reassigned to other facilities. Specific trucks (or the truck heads of trailer trucks) are also normally assigned to drivers who routinely drive them. Such consistency produces efficiency payoffs in terms of vehicle maintenance, payment of drivers, and managing driver workloads. Given the benefits of a larger fleet and more consistent delivery assignments, a distribution center is expected to provide a shorter lead time for retail orders than a plant.

Hypothesis 2. *Compared with the single-DC distribution channel, the factory-direct distribution channel provides a higher fill rate but a longer lead time for retail orders.*

4.2. Moderating Effects of Demand Variability and Order Quantity

In order to satisfy customers' heterogeneous preferences, companies provide a variety of products (Ramdas

2003, Ton and Raman 2010, Tan et al. 2017). The demand variability varies across these products (Chen and Lee 2009). Demand variability is commonly measured using the coefficient of variation (Swaminathan and Tayur 1998, Randall et al. 2006). A large coefficient of variation indicates a high demand variability. Some products (such as popular products) are subject to large average demand and small variations. It is easy for companies to forecast and prepare inventory for these products that have stable demand. Hence, with adequate inventory of these products, I expect the order fill rate for products with low demand variability to be at a relatively high level in the single-DC distribution. Besides, the order fulfillment of these products is rarely delayed by stockouts. Thus, the order lead time for these products is relatively short.

Some products (such as long-tail products) satisfy consumer preference in small market segments. These products normally have small average demand but large variation (Anderson 2006, Tan et al. 2017). The forecasting accuracy for these products is relatively low (Gaur et al. 2007, Simchi-Levi 2010), and therefore, they are more likely to stock out and have a longer lead time than products with low demand variability (Sanders and Wan 2018). Hence, in the single-DC distribution, products with low demand variability are expected to have a better fill rate and shorter lead time than those with high demand variability. In turn, the better order fulfillment for low-variability products leaves less room for improvement by switching distribution channels. Consequently, I expect dual-DC and factory-direct distribution to improve order fill rate and lead time for high-variability products more so than for low-variability products.

Hypothesis 3. *Higher demand variability amplifies the contribution of the dual-DC and factory-direct distribution channels to retail order fill rate and lead time.*

Because inventory capacity varies across retail stores, the order quantity is largely different from one retail store to another. For example, warehouse clubs maintain relatively high inventories, and their orders include large quantities, whereas convenience stores hold limited inventories and place orders with relatively small quantities. Given the limited inventory at distribution centers, large-quantity orders are harder to fulfill than small-quantity orders (Milner and Kouvelis 2005). Consequently, large orders are likely to have lower fill rates and longer lead time than small orders. Hence, the inventory pooling effect generated by dual-DC and factory-direct distribution can better satisfy large-quantity orders and improve their fill rate and lead time.

Hypothesis 4. *Large order quantity amplifies the contribution of the dual-DC and factory-direct distribution channels to retail order fill rate and lead time.*

5. Data and Variables

5.1. Data

I collected data from two different sources: (1) a proprietary data set of orders from and deliveries to retail stores supplied by a consumer packaged goods manufacturing and distribution company and (2) demographic and economic data from the U.S. Census Bureau.

The data from the production and distribution company include proprietary order information, delivery records, and cost information related to the retail stores supplied by the company for 52 weeks in 2012 in a U.S. state. The company made two switches in distribution channels to some retail stores in the state during the data window: in the 21st week, the company reassigned some retail stores supplied by a distribution center from single-DC to factory-direct distribution, and in the 36th week, the company started using a dual-DC distribution channel to supply some stores that were previously supplied by only one distribution center. The switch to dual-DC distribution involves two distribution centers: a primary and an alternative distribution center added to supply the stores previously served only by the primary distribution center. The change to factory-direct distribution involves an original distribution center and the plant to which retail stores had been reassigned. This plant, three distribution centers, and retail stores included in the channel switches are plotted in Online Figure A1 of Online Appendix A.

The company provides the order and delivery records related to retail stores supplied by the plant and three distribution centers included in the channel switches and retail stores in the state supplied by other distribution centers that were not involved in the distribution switches. This plant supplies all distribution centers in the state. In total, the data set includes 2,361 retail stores, among which 963 retail stores were reassigned to the plant and 231 stores were switched to dual-DC distribution. The remaining 1,167 stores were not included in the distribution changes and were consistently supplied by single-DC distribution.

5.2. Variables

The retail stores are categorized according to their distribution channels into three groups: 231 stores that were switched to the dual-DC distribution channel as the treatment group for dual-DC distribution denoted by a dummy variable $DualDCTreat_i$, 963 stores that were switched to the factory-direct distribution channel as the treatment group for factory-direct distribution denoted by a dummy variable $FDDTreat_i$, and 1,167 stores that were not included in the two switches and supplied by single-DC distribution for the entire study period as the control group.

I obtained order fulfillment measures by combining the two data sets provided by the company: one containing order records from these retail stores and the other containing delivery records to retail stores. The order records include the order ID, SKU ID, retail store ID that placed the order, order quantity, and order date. The delivery data set includes the delivery ID, order ID, SKU ID, delivery origin location (plant or DC ID), delivery retail store ID, street address of the retail store, delivery quantity, order date, delivery date, unit price, and unit cost. I merged order and delivery records by retail store ID–order ID–SKU for order fulfillment. In addition, I collected the average inventory level at distribution centers, and these are matched with fulfillment data by DC ID. The merged data set includes 4,735,273 observations at the retail–order–SKU level containing 2,361 retail stores, 162,345 orders, and 610 SKUs in 52 weeks in 2012.

The main variables are explained in Table 1. The performance measures include $FillRate_{ijkt}$, $LeadTime_{ijkt}$, and $Cost_{ijkt}$. Fill rate indicates the percentage of order quantity that is fulfilled. Lead time is measured by the number of days between the retail order and delivery dates. Cost is measured in dollars allocated by the company to the products in deliveries. The costs include three categories: production, inventory, and transportation. The production costs include the raw material, labor, and overhead manufacturing costs. The inventory costs include inventory holding and depreciation costs. The transportation costs include labor (drivers' and warehouse workers' payments) and gas costs. The company allocates these costs in dollars to a unit of each SKU in a delivery. The switch weeks for two distribution channels are denoted by $DualDCWeek_t$ and $FDDWeek_t$, respectively.

In addition to the order quantity, I generate variables to investigate how the SKU and order features moderate the impacts of distribution channels on order fulfillment. Following previous studies (Swaminathan and Tayur 1998, Randall et al. 2006), demand variability ($Variability_{ik}$) is measured using the coefficient of variation: the ratio of the standard deviation to the mean of order quantity of SKU. A large coefficient of variation indicates a high demand variability for SKU k in orders from retail store i. The variable $OrderQuan_{ijkt}$ is the order quantity of SKU k in order j placed by store i in week t. Other variables are explained in Table 1.

I further collect annual demographic and economic data from the 2010 U.S. Census, including population, average household size, total household number, and mean household income. These variables are matched with order fulfillment data by the zip codes within which retail stores are located. Using the data set, I conduct two comparisons: single- versus dual-DC distribution and single-DC versus factory-direct distribution.

6. Single- vs. Dual-DC Distribution

In this section, I investigate the contributions of the dual-DC distribution channel to order fulfillment as compared with single-DC distribution. I first match the stores in the treatment group for dual-DC distribution with those in the control group by PSM. Then, I develop and estimate three equations for fill rate, lead time, and cost.

6.1. Endogeneity Considerations and Matching Retail Stores

Theoretically, the retail stores should be randomly selected in the experiments to explore the treatment effects. The managers in the company confessed that they did not develop a randomized algorithm for retail store selection. Directly comparing order fulfillment between 231 stores in the treatment group for dual-DC distribution and stores in the control group does not provide an apples-to-apples comparison. When designing the research, I consider the potential endogeneity resulting from selection bias. If the endogeneity exists, changes in order fulfillment of retail stores might be caused by factors other than the distribution channels. For example, if the retail stores that were selected for the switch experienced frequently unfulfilled orders, then the order fulfillment improvement of these stores caused by the dual-DC distribution channel may suffer an upward bias. Furthermore, store features, such as order quantity and product variety, may also affect order fulfillment. These variations across retail stores may generate endogeneity through selection bias.

To remove the potential selection bias, I apply PSM using the nearest-neighbor criterion to pair treated stores with control stores, in which the caliper level is set as 0.05 to add a maximum distance constraint on the distance between matched propensities. I match retail stores according to the average inventory level at the distribution center serving the store, the number of retail stores supplied by the distribution center, average weekly order quantity, average number of SKUs in their weekly orders, and economic and demographic factors (such as population, average household size, total household number, and mean household income) in the zip code within which the retail stores are located. The PSM generated 138 pairs of retail stores for the dual-DC distribution treatment and control groups. The propensity scores are constructed using probit regression, which has the likelihood ratio chi-square statistic of 661.18 with a p-value of 0.0001. The estimated results of the probit regression are reported in Online Table B1.

I plot the weekly average fill rate, lead time, and costs for 138 pairs of stores in two groups in Figure 2. Before the switch to dual-DC distribution, fill rate, lead time, and cost were similar between the stores in the treatment and control groups. After the switch, the fill rate started to increase and lead time started to decrease in the treatment group (the dual-DC

Table 1. Variable Descriptions

Variable	Description
$FillRate_{ijkt}$	The unit fill rate for SKU k in order j placed by store i in week t
$LeadTime_{ijkt}$	The lead time in days for SKU k in order j placed by store i in week t
$Cost_{ijkt}$	The cost in dollars per unit (including production, inventory, and transportation costs) for SKU k in order j placed by store i in week t
$OrderQuan_{ijkt}$	The order quantity in units of SKU k in order j placed by store i in week t
$DeliveryQuan_{ijkt}$	The delivery quantity in units of SKU k in order j placed by store i in week t
$DualDCTreat_i$	A dummy variable for retail stores in the dual-DC distribution channel. It equals one for stores that were switched to dual-DC distribution and zero otherwise.
$DualDCWeek_t$	A dummy variable for the week t after dual-DC distribution is employed to retail stores. It is coded one after week 36 in 2012 and zero otherwise.
$FDDTreat_i$	A dummy variable for retail stores in the factory-direct distribution channel. It equals one for stores that were switched to factory-direct distribution and zero otherwise.
$FDDWeek_t$	A dummy variable for the week t after factory-direct distribution is employed to retail stores. It is coded one after week 21 in 2012 and zero otherwise.
$Variability_{ik}$	The demand variability for SKU k in orders from store i. It is measured by the coefficient of variation of order quantity, which is the ratio of the standard deviation to the average order quantity.
$Price_{ijkt}$	The price in dollars for a unit of SKU k in order j placed by store i in week t
$DCInv_{ikt}$	The average inventory of SKU k in the distribution center supplying store i in week t
$DCStore_{it}$	The number of retail stores served by the distribution center supplying store i in week t
$OrderPV_{ijt}$	The number of SKUs included in order j placed by store i in week t
$OrderFreq_{it}$	The number of orders placed by store i in week t
$DeliveryFreq_{it}$	The number of deliveries to store i in week t
$Population_i$	The population in the zip code within which store i is located
$HouseholdNum_i$	The total household number in the zip code within which store i is located
$HouseholdIncome_i$	The mean household income in the zip code within which store i is located
$HouseholdSize_i$	The average household size in the zip code within which store i is located

distribution group). Meanwhile, the plot of costs demonstrates a clear seasonality over time but does not show any clear change in the difference between the two groups. These plots offer some model-free evidence for the parallel trend assumption of the DiD analysis. I statistically test the trend in these three variables in Section 6.4. The seasonal patterns in these three variables suggest the need for including fixed effects for time in the empirical model.

The statistics for order fulfillment and cost of the matched retail stores before the switch are summarized in Online Table B2. The two-tailed t-tests are conducted to compare these variables in the treatment and control groups. None of the tests reject the null hypothesis of zero difference, which suggests no significant difference between these two groups. Online Table B2 suggests that the PSM generates matched samples for the empirical analysis. The data set with the matched retail stores contains 517,599 observations at the store–order–SKU level, 486 SKUs, and 16,727 orders placed by 276 retail stores. In addition to PSM that addresses the selection bias on the observable variables, I applied SCM (Abadie et al. 2010). The steps and results of SCM are explained in Section 6.4.

6.2. Empirical Model

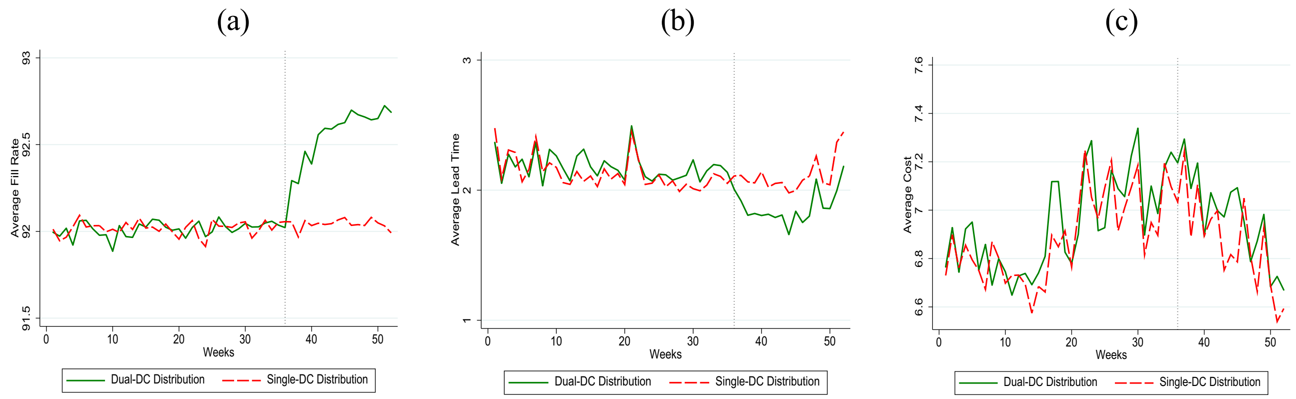
I develop three equations to investigate the influence of the switch from single-DC to dual-DC distribution, in which Equation (1) is constructed for fill rates, Equation (2) represents lead time, and Equation (3) represents costs. Because the changes in distribution channels are associated with changes in cost, I include the cost equation to study the trade-offs between order fulfillment and cost resulting from the distribution channels. The natural logarithmic transformation is applied to the dependent variables and some control variables in three equations for a closer approximation of a normal distribution. Errors in each equation are clustered by retail stores.

$$\begin{aligned}\log FillRate_{ijkt} = {} & store_i + week_t + \alpha_1 DualDCTreat_i \\ & \times DualDCWeek_t + \alpha_2 \log Variability_{ik} \\ & + \alpha_3 \log Price_{ijkt} + \alpha_4 \log OrderQuan_{ijkt} \\ & + \alpha_5 \log OrderPV_{ijt} + \varepsilon_{ijkt},\end{aligned} \tag{1}$$

$$\begin{aligned}\log LeadTime_{ijkt} = {} & store_i + week_t + \beta_1 DualDCTreat_i \\ & \times DualDCWeek_t + \beta_2 \log Variability_{ik} \\ & + \beta_3 \log Price_{ijkt} + \beta_4 \log OrderQuan_{ijkt} \\ & + \beta_5 \log OrderPV_{ijt} + e_{ijkt},\end{aligned} \tag{2}$$

$$\begin{aligned}\log Cost_{ijkt} = {} & store_i + week_t + \gamma_1 DualDCTreat_i \\ & \times DualDCWeek_t + \gamma_2 \log Variability_{ik} \\ & + \gamma_3 \log Price_{ijkt} + \gamma_4 \log OrderQuan_{ijkt} \\ & + \gamma_5 \log OrderPV_{ijt} + \xi_{ijkt}.\end{aligned} \tag{3}$$

Figure 2. (Color online) Dual-DC vs. Single-DC Distribution: Fill Rate, Lead Time, and Cost



Notes. (a) Fill rate. (b) Lead time. (c) Cost.

In all three equations, the fixed effects for retail stores and weeks ($store_i$ and $week_t$) are included for unobservable variations across stores and over weeks. I estimate the three equations using least square regression with dummy variables to capture fixed effects. The dummy variable $DualDCTreat_i$ indicates stores in the treatment group for dual-DC distribution. $DualDCWeek_t$ indicates the weeks after dual-DC distribution is used. When the fixed effects for stores and weeks are included in the model, the main effects of $DualDCTreat_i$ and $DualDCWeek_t$ cannot be identified separately from these fixed effects. Hence, I include only the interaction term $DualDCTreat_i \times DualDCWeek_t$ in the model when these fixed effects are included. The coefficient of this interaction term captures the treatment effect of dual-DC distribution. Each equation is developed using a log-linear regression expression, in which these three coefficients indicate the percentage changes in order fulfillment and cost as a result of the switch to dual-DC distribution. In addition to the main variable of interest, I include control variables in each equation to control for their influence on order fulfillment and cost: demand variability ($Variability_{ik}$), order quantity ($OrderQuan_{ijkt}$), price ($Price_{ijkt}$), and the number of SKUs in orders ($OrderPV_{ijt}$).

The descriptive statistics of variables included in the matched data set for the dual-DC distribution channel are summarized in Table 2. These stores have a decent order fulfillment performance. Fill rate ($FillRate_{ijkt}$) has a mean of 92.082. The order fulfillment lead time ($LeadTime_{ijkt}$) is two days on average. The cost per unit ($Cost_{ijkt}$) has a mean value of 6.944 dollars.

6.3. Treatment Effect of Dual-DC Distribution

The estimated results of Equations (1)–(3) are reported in Table 3. The results of Equation (1) for fill rate are reported in the first column. The treatment effect is indicated by the estimated coefficient of $DualDCTreat_i \times DualDCWeek_t$, which is positive and significant. This indicates an increase in fill rates when dual-DC distribution is employed. After switching to dual-DC distribution, the fill rate for retail orders increases by 0.4%. Because the average fill rate is already at a high level with a mean value of 92%, the increase in fill rate seems relatively small. The improved fill rates result from the pooling effect generated by the alternative distribution center.

As shown in the second column of Table 3, the negative treatment effect on lead time suggests a 9.7% decrease in lead time when the dual-DC distribution is employed. The alternative distribution center increases the supply flexibility to retail orders. When the primary distribution center does not have enough inventory for retail orders, instead of waiting, the retail orders can be fulfilled by the alternative distribution center. The positive treatment effect of dual-DC distribution on the fill rate and the negative treatment effect on lead time support Hypothesis 1.

I report the estimated results of Equation (3) for the cost in the last column of Table 3. The treatment effect of dual-DC distribution on cost is not significant. That is, there is no significant difference in cost between single-DC and dual-DC distribution. This result may be explained by the fact that there is no retail order fulfilled simultaneously by both distribution centers. Even in dual-DC distribution, either the primary or the alternative distribution center fulfills a retail order. The cost associated with the delivery to stores from the alternative distribution center is similar to the cost associated with the delivery from the primary distribution center.

For control variables, higher demand variability is associated with lower fill rate, longer lead time, and higher costs, which is consistent with the literature that finds demand variability amplifies the mismatch

Table 2. Descriptive Statistics for Variables Used for Single-DC vs. Dual-DC Distribution

Variable	Unit	Mean	Standard deviation
$FillRate_{ijkt}$	Percentage	92.082	2.005
$LeadTime_{ijkt}$	Day	2.099	0.866
$Cost_{ijkt}$	Dollar	6.944	3.302
$DualDCTreat_i$	Zero or one	0.486	0.500
$DualDCWeek_t$	Zero or one	0.309	0.462
$Variability_{ik}$	Ratio	0.505	0.375
$Price_{ijkt}$	Dollar	13.159	5.829
$OrderQuan_{ijkt}$	Unit	4.422	17.638
$OrderPVijt$	SKU	32.984	21.257
$DeliveryQuan_{ijkt}$	Unit	4.018	16.594
$OrderFreq_{it}$	Count	1.610	0.876
$DeliveryFreq_{it}$	Count	1.609	0.865

between supply and demand (Chen and Lee 2009, Simchi-Levi 2010). Order quantity is negatively associated with fill rate and cost but positively related to lead time. Orders with large quantities are more likely not being fully filled because of the limited inventory at distribution centers. Because of the economies of scale in production and transportation, large order quantities result in lower average costs. Price is negatively associated with lead time but positively related to fill rate and cost. On average, higher product variety in an order is associated with a lower fill rate, a longer lead time, and a higher cost. These results are consistent with the existing literature on product variety (Lancaster 1990, Wan et al. 2012).

6.4. Robustness Check

6.4.1. Synthetic Control Method. In the main models, I match the retail stores in the treatment and control groups using PSM and then apply the DiD method. Other than using PSM, I also apply the synthetic control method (Abadie et al. 2010, 2015) as a robustness check. Compared with PSM, the synthetic control method exploits all stores in the control group and constructs one synthetic store that best approximates each treated store (Abadie et al. 2010).

Although the synthetic control method is normally used to study scenarios with only one treated unit (Abadie et al. 2010, 2015; Tirunillai and Tellis 2017; Lu et al. 2021), I extend this method to multiple stores in the treatment group. Specifically, I construct one synthetic store using stores in the control group for each of the 231 stores in the treatment group. Fill rate, lead time, and cost are used as the outcome variable, and the following variables are used as the pretreatment characteristics to construct the synthetic store: weekly inventory level; the number of retail stores supplied by the distribution center serving the stores; weekly order quantity; the average number of SKUs in the weekly orders of retail stores; and population, average household size, total household number, and mean

Table 3. Treatment Effect of the Dual-DC Distribution Channel on Fill Rate, Lead Time, and Cost

Variable	$\log FillRate_{ijkt}$	$\log LeadTime_{ijkt}$	$\log Cost_{ijkt}$
$DualDCTreat_i \times DualDCWeek_t$	0.004***	−0.097***	0.005
	(0.001)	(0.006)	(0.004)
$Variability_{ik}$	−0.002***	0.012***	0.074***
	(0.000)	(0.003)	(0.013)
$\log Price_{ijkt}$	0.001***	−0.007**	0.496***
	(0.000)	(0.002)	(0.012)
$\log OrderQuan_{ijkt}$	−0.002***	0.010***	−0.043***
	(0.000)	(0.002)	(0.003)
$\log OrderPV_{ijt}$	−0.002*	0.031**	0.007**
	(0.000)	(0.011)	(0.002)
Observations	517,599	517,599	517,599
Fixed effects for retail sores	Included	Included	Included
Fixed effects for year–week	Included	Included	Included
R^2	0.524	0.664	0.441
Adjusted R^2	0.466	0.663	0.441

***$p < 0.001$; **$p < 0.01$; *$p < 0.05$.

household income in the zip code within which the retail stores are located. The details of the synthetic control method are reported in Online Appendix C, in which the estimated results are consistent with those reported in Table 3 using the PSM method.

6.4.2. Treatment Effect with No Matching. The main results in Table 3 are estimated using the data of matched treated stores and control stores selected by PSM. To ensure that the results remain robust without applying any matching method, I estimate Equations (1)–(3) again using the data set before matching. The estimated treatment effects are reported in Online Table D1. The estimated treatment effects on fill rate, lead time, and cost are consistent with the main results using PSM but have less magnitude.

6.4.3. Treatment Effect with One-to-Many Matching and Different Caliper Level. In the main results, the models are estimated using the matched data sample by one-to-one PSM with a caliper level of 0.05. To ensure that the findings are not limited by the caliper level and one-to-one matching, I allow one-to-many matching, which generates 138 treated stores and 619 control stores for the dual-DC distribution. The estimated results using the data set by the one-to-many matching are reported in Online Table E1. Besides, I increase the caliper level to 0.1 in PSM, which generates 163 pairs of treated and control stores. Then, I estimate Equations (1)–(3) using this data set. The estimated results about the treatment effects in Online Table E1are consistent with the main results.

6.4.4. Time Trends. I examine whether the changes in order fulfillment and cost can be explained by time trends that may have existed prior to the switch to dual-DC distribution. That is, I rule out the possibility that the changes in order fulfillment and cost result from some consistent time trends in the weeks before the switch to dual-DC distribution. The models and estimations are reported in Online Appendix F. These estimated coefficients in Online Table F1 suggest that there are parallel time trends of fill rates, lead time, and cost between the treatment and control groups before the switch to the dual-DC distribution channel.

6.4.5. Placebo Tests. Finally, I conduct placebo tests on the models by pretending that dual-DC distribution started in some other weeks before the actual switch to the dual-DC distribution channel. Then, I estimate Equations (1)–(3) using these placebo dual-DC distribution weeks by replacing $DualDCTreat_i \times DualDCWeek_t$ with $DualDCTreat_i \times PlaceboWeek_t$. The estimated coefficients are reported in Online Table G1 and show that none of these estimated coefficients for placebo weeks is significant. In sum, all of the aforementioned robustness tests suggest that fill rate increases, lead time reduces, and costs remain unchanged after the dual-DC distribution channel is used.

7. Single-DC vs. Factory-Direct Distribution

In this section, I analyze the trade-offs between order fulfillment and the cost associated with the factory-direct distribution channel.

7.1. Matching Retail Stores and Empirical Models

I applied PSM with a caliper level of 0.05 to pair the 963 retail stores in the treatment group for factory-direct distribution and the retail stores in the control group and use the following matching criteria: the average inventory level and number of retail stores supplied by the distribution center serving the stores, average weekly order quantity, average number of SKUs in the weekly order, population, average household size, total household number, and mean household income. PSM finds 646 pairs of stores. The probit regression used in propensity score calculation yields the likelihood ratio chi-square statistic of 220.13 with a p-value of 0.0001. The estimated results are reported in Online Table B1.

The statistics for the order features of the matched retail stores and the t-test results are summarized in Online Table B3. The data set with the matched treatment and control groups contains 646 pairs of retail stores, 93,725 orders, 603 SKUs, and 2,764,066 observations in total. The descriptive statistics of variables included in the data set are summarized in Table 4. I plot the average fill rate, lead time, and cost in the matched treatment and control groups for the factory-direct distribution channel in Figure 3. Before the switch to factory-direct distribution (week 21), fill rate, lead time, and cost are similar between the treatment and control groups. After the switch, fill rate and lead time increase but cost decreases in the treatment group. I develop three equations for fill rate, lead time, and cost by replacing $DualDCTreat_i \times DualDCWeek_t$ in Equations (1)–(3) with $FDDTreat_i \times FDDWeek_t$ to capture the treatment effect of the factory-direct distribution.

7.2. Treatment Effect of Factory-Direct Distribution

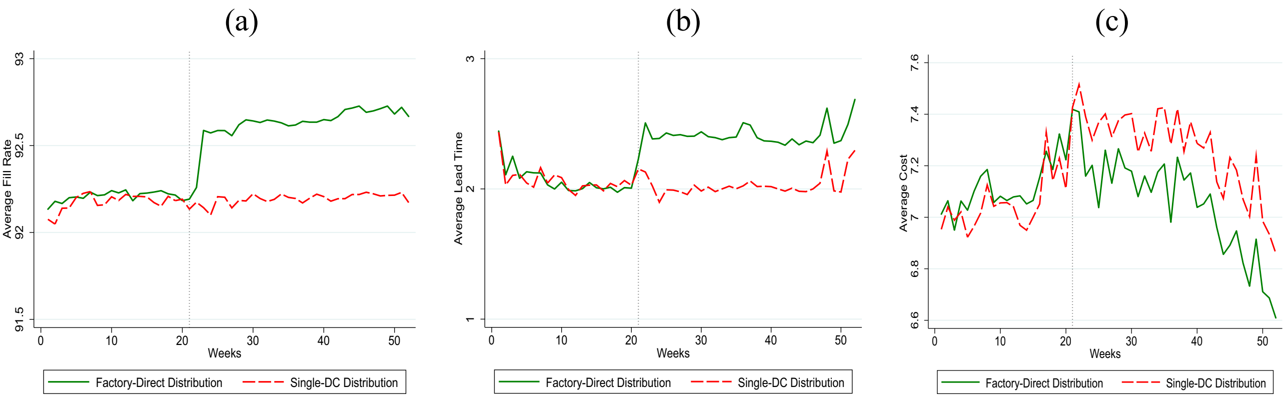
The estimated results are reported in Table 5. These results suggest that factory-direct distribution increases retail order fill rate by 0.5%, prolongs lead time by 12.5%, and reduces costs by 5.2%. Hypothesis 2 is supported. The increases in fill rate come from the large supply capability of the plant, which reduces the unfulfilled retail order quantity. On the other hand, as a consequence of a lower delivery frequency and limited inventory personnel and distribution fleets allocated for direct delivery to retail stores, the order processing and delivery time at a plant is longer than at distribution centers, which leads to a longer lead time. I investigate the changes in delivery frequency after the switch to factory-direct distribution in Section 8.2.

Cost-savings are achieved by skipping the distribution center. Before the switch to factory-direct distribution, products were delivered from the plant to the distribution center and were stored there while awaiting retail orders. When retail orders arrived, they were delivered from the distribution center to the retail stores. After the switch, the transportation costs associated with the deliveries from the distribution center to stores are reduced, especially when the plant's location is not significantly farther from the stores than the distribution center as shown in Online Figure A1. Moreover, transportation costs can be further reduced by the low delivery frequency to stores in factory-direct distribution. The results of control

Table 4. Descriptive Statistics for Variables Used for Single-DC vs. Factory-Direct Distribution

Variables	Unit	Mean	Standard deviation
$FillRate_{ijkt}$	Percentage	92.247	2.252
$LeadTime_{ijkt}$	Day	2.164	0.854
$Cost_{ijkt}$	Dollar	7.135	3.498
$FDDTreat_i$	Zero or one	0.512	0.500
$FDDWeek_t$	Zero or one	0.607	0.488
$Variability_{ik}$	Ratio	0.422	0.349
$Price_{ijkt}$	Dollar	13.531	5.889
$OrderQuan_{ijkt}$	Unit	3.868	15.877
$OrderPV_{ijt}$	SKU	33.884	24.440
$DeliveryQuan_{ijkt}$	Unit	3.588	15.864
$OrderFreq_{it}$	Count	1.678	0.900
$DeliveryFreq_{it}$	Count	1.665	0.956

Figure 3. (Color online) Factory-Direct vs. Single-DC Distribution: Fill Rate, Lead Time, and Cost



Notes. (a) Fill rate. (b) Lead time. (c) Cost.

variables are consistent with those in Table 3. I also conduct robustness checks similar to those for dual-DC distribution. The results are reported in Online Appendices C–G, which are consistent with the main results in Table 5.

8. Mechanism of Dual-DC and Factory-Direct Distribution

A change in order fulfillment may be a result of changes on the demand and/or the supply side because of a switch in distribution channels. For example, the increases in fill rate caused by dual-DC and factory-direct distributions may result from (1) decreases in retail orders, (2) increases in delivery capability, or (3) both. A decrease in delivery frequency along with an increase in delivery quantity may lead to an increase in lead time and a drop in cost. Hence, I explore the mechanism of changes in order fulfillment and cost resulting from distribution channels by analyzing their influences on retail orders and deliveries.

Table 5. Treatment Effect of Factory-Direct Distribution on Fill Rate, Lead Time, and Cost

Variables	log$FillRate_{ijkt}$	log$LeadTime_{ijkt}$	log$Cost_{ijkt}$
$FDDTreat_i \times FDDWeek_t$	0.005***	0.125***	−0.052***
	(0.000)	(0.004)	(0.002)
$Variability_{ik}$	−0.003***	0.015***	0.096***
	(0.000)	(0.002)	(0.007)
log$Price_{ijkt}$	0.001***	−0.010***	0.576***
	(0.000)	(0.001)	(0.005)
log$OrderQuan_{ijkt}$	−0.003***	0.006***	−0.049***
	(0.000)	(0.001)	(0.002)
log$OrderPV_{ijt}$	−0.001***	0.045***	0.008***
	(0.000)	(0.005)	(0.002)
Observations	2,764,066	2,764,066	2,764,066
Fixed effects for retail sores	Included	Included	Included
Fixed effects for year–week	Included	Included	Included
R^2	0.489	0.628	0.472
Adjusted R^2	0.483	0.627	0.472

***$p < 0.001$; **$p < 0.01$; *$p < 0.05$.

8.1. Retail Order Quantity and Delivery Quantity

I investigate whether the retail order quantity changes after the switches to dual-DC distribution and factory-direct distribution. The results are reported in Online Table H1. The order quantity does not change significantly in response to dual-DC and factory-direct distributions. It is likely that the retail stores had not realized or responded to the change in distribution channels during the time window in the data.

Because the retail order quantity has not changed, the contributions of dual-DC and factory-direct distributions to order fulfillment should come from the changes in deliveries. I examine the delivery quantity using DiD models. The estimated results are reported in Online Table H1. After switching to the dual-DC and factory-direct distribution channels, retail stores receive larger delivery quantities. These results demonstrate that the dual-DC and factory-direct distribution channels raise the fill rates of retail orders by increasing delivery quantities.

8.2. Order Frequency and Delivery Frequency

Whereas the results of order and delivery quantity explain the improvement in fill rate, I believe that changes in lead time can be explained by order and delivery frequency. In an ideal case in which a distribution center always has adequate inventory, it should arrange a delivery to a retail store after receiving an order from it. Then, the delivery frequency would be the same as the order frequency. However, when stockouts occur in this distribution center, it cannot deliver products to fulfill the retail order in time. Some of these unsatisfied orders can be aggregated to this retail store's future orders. As a result, these multiple orders from the retail store are fulfilled by one delivery. When order frequency is unchanged, decreases in the delivery frequency suggest more order aggregations and a longer lead time. To investigate order and delivery frequency, I aggregate the data to the store–week level observations, where $OrderFreq_{it}$ denotes the number of orders placed by store i in week t and $DeliveryFreq_{it}$ denotes the number of deliveries to store i in week t.

The estimated treatment effects on order and delivery frequency are reported in Online Table H2. The order frequency from retail stores does not significantly change after the switches to the dual-DC and factory-direct distribution channels. However, the delivery frequency decreases after the switch to the factory-DC distribution channel but does not change after the switch to the dual-DC distribution channel. These results explain the longer lead time after the switch to the factory-direct distribution channel.

In sum, I find that, whereas retail order quantity does not change, the delivery quantity increases after the switches to the dual-DC and factory-direct distribution channels. Compared with single-DC distribution, dual-DC and factory-direct distribution provide stronger supply capabilities, which results in a higher fill rate. Furthermore, the results suggest that delivery frequency reduces after the switch to factory-direct distribution. The less frequent deliveries explain the longer lead time associated with the factory-direct distribution as the plant aggregates more orders in a delivery. The larger delivery quantities and smaller delivery frequencies jointly generate the economy of scale in deliveries, which contributes to the lower costs associated with factory-direct distribution. Although the delivery frequency and cost do not change after the switch to dual-DC distribution, the inventory availability at the backup distribution center enables the shorter lead time in the dual-DC distribution channel.

9. Order Fulfillment Improvement Across Demand Variability and Order Quantity

To better understand the contributions of these distribution channels, I explore how their treatment effects vary across demand variability and order quantity. Specifically, I modify the main models by adding $DualDCTreat_i \times DualDCWeek_t \times Variability_{ik}$ and $DualDC\ Treat_i \times Dual\ \ DCWeek_t \times \log OrderQuan_{ijkt}$, respectively, to the equation for the dual-DC distribution channel and then adding $FDDTreat_i \times FDDWeek_t \times\ Variability_{ik}$ and $FDD\ Treat_i \times FDDWeek_t \times \log Order\ Quan_{ijkt}$ to the equation for the factory-direct distribution channel. Their coefficients suggest the moderating effects of demand variability and order quantity. The estimated coefficients for these DDD terms are summarized in Table 6.

These results suggest that, after the switch to dual-DC distribution, products with high demand variability receive improvements on fill rate and lead time more than products with low demand variability. Meanwhile, after the switch to factory-direct distribution, products with high demand variability have fill rate increases and cost decreases more than products with low demand variability. The improved fill rates for high-variability products result from the inventory-pooling effect, which contributes more to the mismatch between supply and demand caused by high-variability products. The high-variability products do not have lower costs than low-variability products after the switch to the dual-DC distribution or shorter lead time after the switch to the factory-direct distribution. This is because, as shown in Tables 3 and 5, dual-DC distribution has no cost advantage and factory-direct distribution has no lead time advantage over single-DC distribution. On average, these results support Hypothesis 3: the order fulfillment improvement by dual-DC and factory-direct distribution is larger for products with higher demand variability.

The results in Table 6 also support Hypothesis 4. Compared with the single-DC distribution, the fill rate increases and lead time decreases for large-quantity orders more than small-quantity orders after the switch to dual-DC distribution. Moreover, the fill rate is increased, and cost is reduced for large-quantity orders by factory-direct distribution more so than for small-quantity orders. The larger fill rate improvements of distribution channel switches for large-quantity orders indicate that orders with large quantities benefit more from the increased supply capacity in the dual-DC and factory-direct distribution. Cost-saving by the factory-direct distribution channel is amplified by the economies of scale associated with large order quantities.

10. Managerial Implications

I provide analyses of changes in order fulfillment and cost after a manufacturing and distribution company switches its retail stores to the dual-DC and factory-direct distribution channels from the traditional single-DC distribution channel. The findings offer guidance to companies employing different distribution channels for their retail customers. The results suggest that, compared with traditional single-DC distribution, the dual-DC distribution channel helps raise the fill rate and reduce the lead time of retail orders without a sacrifice of production and distribution cost, whereas the factory-direct distribution channel contributes to higher fill rate and cost savings yet longer lead times. In summary, when dual-DC distribution is used, the fill rate increases by 0.4%, the lead

Table 6. Moderating Effects of Demand Variability and Order Quantity

	Dual-DC distribution		
Variable	$\log FillRate_{ijkt}$	$\log LeadTime_{ijkt}$	$\log Cost_{ijkt}$
$DualDCTreat_i \times DualDCWeek_t \times Variability_{ik}$	0.003**	−0.007**	−0.014
	(0.001)	(0.002)	(0.013)
$DualDCTreat_i \times DualDCWeek_t \times \log OrderQuan_{ijkt}$	0.002*	−0.003**	−0.001
	(0.001)	(0.001)	(0.004)
	Factory-Direct distribution		
Variable	$\log FillRate_{ijkt}$	$\log LeadTime_{ijkt}$	$\log Cost_{ijkt}$
$FDDTreat_i \times FDDWeek_t \times Variability_{ik}$	0.005*	0.001	−0.030***
	(0.002)	(0.004)	(0.009)
$FDDTreat_i \times FDDWeek_t \times \log OrderQuan_{ijkt}$	0.002*	0.000	−0.012***
	(0.001)	(0.001)	(0.002)

Note. The fixed effects and all control variables are included in the estimations.
***$p < 0.001$; **$p < 0.01$; *$p < 0.05$.

time reduces by 9.7%, and costs remain unchanged. Meanwhile, the factory-direct distribution raises the fill rate by 0.5%, saves cost by 5.2%, but increases the lead time by 12.5%.

In the age of omnichannel distribution, it is important for managers to understand the trade-offs associated with different distribution channels. Before selecting a distribution channel, companies should evaluate the potential trade-offs associated with each distribution channel option. For companies that offer a relatively high fill rate and plan to reduce lead time for better customer service, a dual-DC distribution channel is a good option. Meanwhile, the companies, which exclusively own the sales and distribution channel of branded products to retail stores, may put more weight on costs. A factory-direct distribution channel fits their needs by reducing distribution costs.

Managers also need to understand the multiple dimensions of order fulfillment when making decisions on distribution channels. Although various distribution channels have the potential to improve retail order fulfillment, these distribution channels should be evaluated by different fulfillment dimensions. The fill rate indicates product availability, whereas the lead time reveals the speed of order fulfillment. These order fulfillment dimensions may not always change in one direction. For example, when switching to factory-direct distribution, fill rate increases but lead time is prolonged.

Moreover, the treatment effects of dual-DC and factory-direct distribution on order fulfillment are contingent on demand variability at the SKU level. A company obtains greater fulfillment improvement when switching products with a higher demand variability from a single-DC distribution channel to a dual-DC or factory-direct channel. This finding is aligned with the literature suggesting the channel choices should consider product selections (Brynjolfsson et al. 2009). Therefore, it could be beneficial for companies to allocate products with different features to different distribution channels.

I also find that the contributions of distribution channels to order fulfillment are affected by retail order quantity. In general, a larger order quantity benefits more from dual-DC and factory-direct distribution channels on fill rate enhancement, lead time, and cost reduction. Because order quantity varies largely across retailers, companies may consider allocating various distribution channels to supply different retail stores based on their order features. Furthermore, whereas independent retailers mainly rely on suppliers' distribution channels to fulfill their orders, retail chains may develop their own distribution channels. The findings provide insights to retail chains when they build their distribution channels. Stores in supermarket chains place order quantities larger than stores in convenience store chains. When developing new distribution channels, supermarket chains are more likely to choose dual-DC and factory-direct distribution channels than convenience store chains because the fulfillment of their large order quantity benefits more from these two channels than the small order quantities by convenience stores.

11. Conclusions and Limitations

It is important to understand the contribution of various distribution channels to order fulfillment as well as the trade-offs between improved order fulfillment and costs. This paper designs a series of DiD analyses and exploits two distribution channel switches. By empirically investigating these two events, I confirm and quantify the value of the dual-DC and factory-direct distribution channels to the fill rate and lead time of retail orders as well as their cost consequences.

The results enrich the understanding of the trade-offs between order fulfillment and cost associated with distribution channels. The findings provide guidance to production and distribution companies interested in implementing new distribution channels. However, this study suffers a few limitations. For example, the identification strategy relies on two events. This paper does not explore other distribution channels in supply chains. Future research can empirically examine the contribution of more distribution channels when more data are available.

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